

2024 HKDSE Physics

Paper 1A — Multiple-choice Questions

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

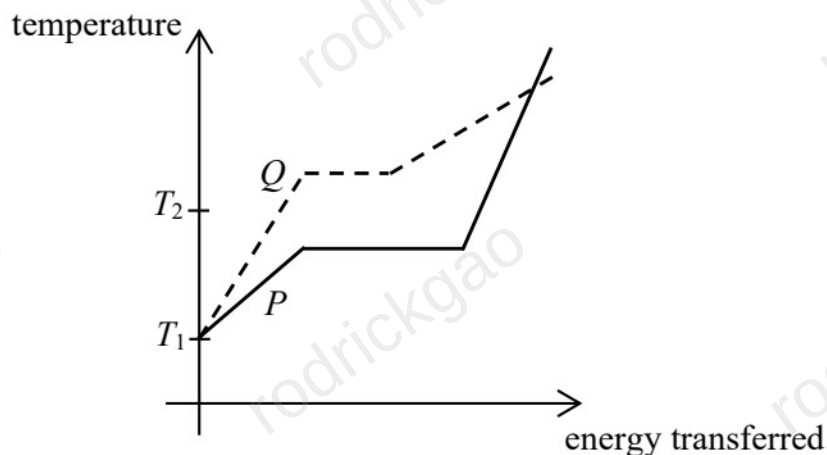
Questions

Official answers are placed directly below each question.

Question 1

Substances P and Q are of the same mass and both are solids at temperature T_1 . The graph below shows the variation of temperature of the substances with energy transferred.

Which statement below is correct?



- A. P has a higher melting point than Q.
- B. The specific latent heat of fusion of P is smaller than that of Q.
- C. The specific heat capacity of solid P is greater than that of solid Q.
- D. At temperature T_2 , P is gas while Q is solid.

Official answer: C

For equal masses, the slope of a temperature–energy graph is $\Delta T/Q = 1/(mc)$. Solid P has the smaller initial slope, so its specific heat capacity is greater. Its plateau is lower and longer: P has a lower melting point and a larger latent heat of fusion. At T_2 , P is liquid, not gas, while Q is still solid. Answer: C.

Question 2

If 0.5 kg of water at 50 °C is mixed with 2.0 kg of ice at 0 °C, what is the final temperature of the mixture?

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹ specific latent heat of fusion of ice = 3.34×10^5 J kg⁻¹

- A. -54 °C
- B. 0 °C
- C. 10 °C
- D. 25 °C

Official answer: B

Cooling the water to 0 °C releases $Q = mc\Delta T = 0.5 \times 4200 \times 50 = 1.05 \times 10^5$ J. Melting all the ice requires $ml = 2.0 \times 3.34 \times 10^5 = 6.68 \times 10^5$ J. Only 0.314 kg of ice melts, so ice and water coexist at equilibrium and the final temperature is 0 °C. Answer: B.

Question 3

The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm³ of such 'vacuum' at room temperature.

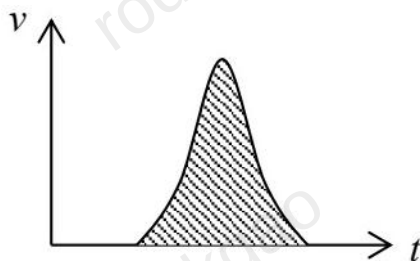
- A. 10^4
- B. 10^6
- C. 10^8
- D. 10^{12}

Official answer: B

Use $pV = NkT$, with $V = 1 \text{ cm}^3 = 10^{-6} \text{ m}^3$ and $T \approx 300 \text{ K}$. $N = (10^{-8} \times 10^{-6}) / (1.38 \times 10^{-23} \times 300) \approx 2.4 \times 10^6$ molecules. Thus the order of magnitude is 10^6 . A very low pressure does not imply an empty volume. Answer: B.

Question 4

What physical quantity does the shaded area represent in the above velocity-time (v-t) graph?



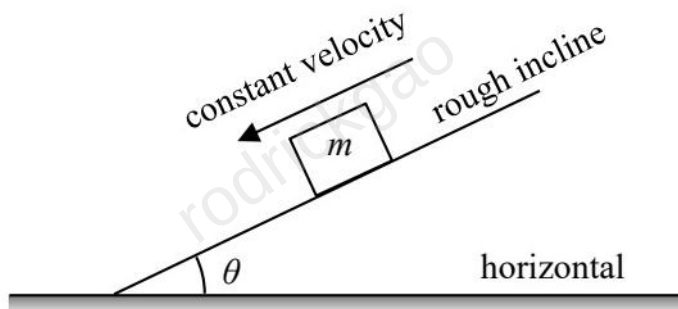
- A. acceleration
- B. change of velocity
- C. momentum
- D. displacement**

Official answer: D

Displacement is the signed area under a velocity-time graph: $\Delta s = \int v \, dt$. The units are $(\text{m s}^{-1}) \times \text{s} = \text{m}$. The slope, rather than the area, gives acceleration. Answer: D.

Question 5

A block of mass m slides down along a fixed rough incline with constant velocity as shown. The incline makes an angle θ with the horizontal. What is the magnitude of the force exerted on the incline by the block?



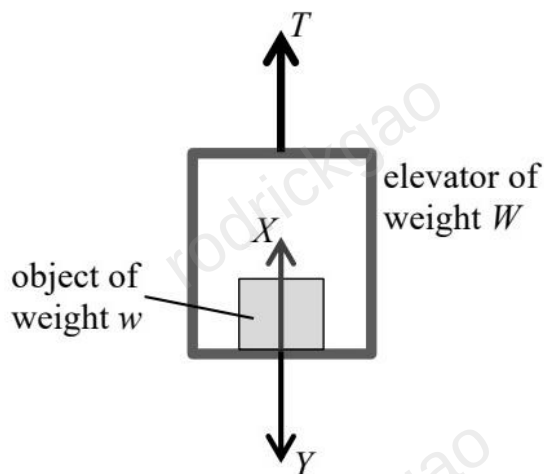
- A. zero
- B. mg**
- C. $mg \sin \theta$
- D. $mg \cos \theta$

Official answer: B

Constant velocity means zero resultant force on the block. Its weight is balanced by the combined normal reaction and friction from the incline, so the resultant contact force on the block is vertically upward with magnitude mg . By Newton's third law, the block exerts an equal downward force mg on the incline. The question asks for the total contact force, not just $mg \cos \theta$. Answer: B.

Question 6

An object of weight w is placed inside an elevator of weight W . A force T pulls the elevator upward. A supporting force X acts on the object while the floor of the elevator experiences a force Y due to the presence of the object. Which of the following is an action and reaction pair?



- A. T and $(W + w)$
- B. w and X
- C. X and Y
- D. It cannot be determined as the direction of acceleration of the elevator is unknown.

Official answer: C

X is the force of the floor on the object, and Y is the force of the object on the floor. They are equal and opposite forces from the same interaction acting on different bodies, regardless of the elevator's acceleration. Weight w and support X both act on the object, so they are not an action–reaction pair. Answer: C.

Question 7

In an experiment, a horizontal force F acts on a block placed on a rough horizontal surface as shown in Figure (a). Figure (b) shows the variation of the block's kinetic energy E_k with the distance s it travelled. Neglect air resistance and assume frictional force unchanged.

If the experiment is repeated with force F doubled, which of the following graphs represents the expected result (in dotted line)?

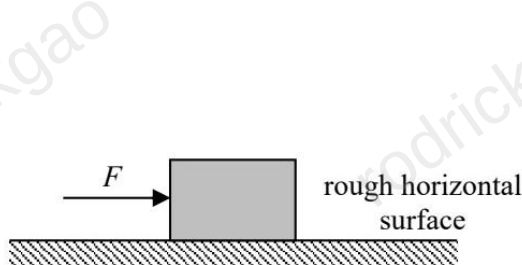


Figure (a)

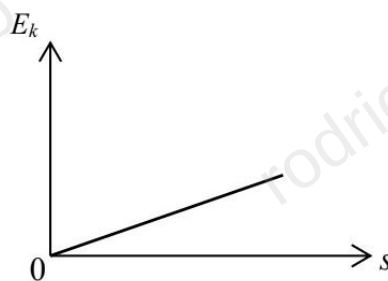
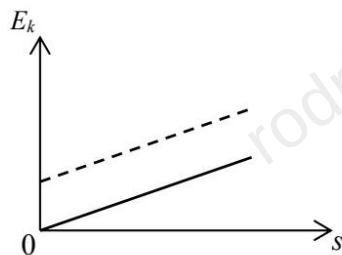
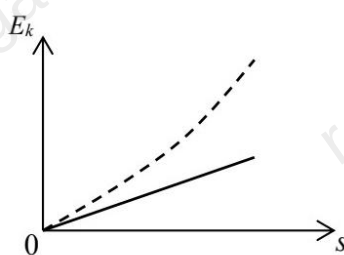


Figure (b)

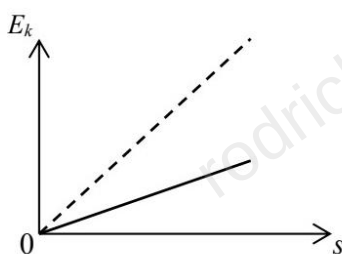
A.



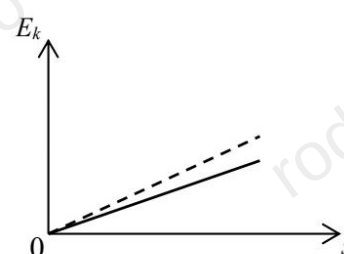
B.



C.



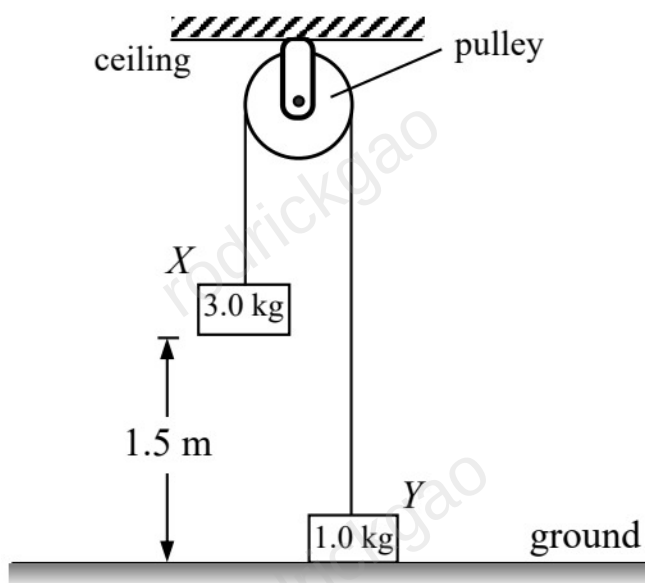
D.

**Official answer: C**

The work–energy theorem gives $E_k = (F - f)s$, where f is the unchanged friction. Doubling F gives $E_k = (2F - f)s$: a straight line through the origin. Its slope is more than twice the original slope because $2F - f > 2(F - f)$. This is the steeper straight dotted line in C. Answer: C.

Question 8

Block X and Y of mass 3.0 kg and 1.0 kg respectively are connected by a light inextensible string passing over a smooth light pulley fixed to the ceiling as shown. The blocks are released from rest. Find the time taken for block X to reach the ground. Neglect air resistance.



- A. 0.55 s
- B. 0.64 s
- C. 0.68 s
- D. 0.78 s

Official answer: D

Treat both blocks as one system: $a = (3.0 - 1.0)g / (3.0 + 1.0) = 4.905 \text{ m s}^{-2}$. X starts from rest and falls 1.5 m, so $s = \frac{1}{2}at^2$ gives $t = \sqrt{(2 \times 1.5 / 4.905)} = 0.782 \text{ s} \approx 0.78 \text{ s}$. The acceleration is less than g because the string supports X. Answer: D.

Question 9

A person thought that when he pulls his hair with a large enough force, eventually he will be able to pull himself off the ground. This is a misconception because

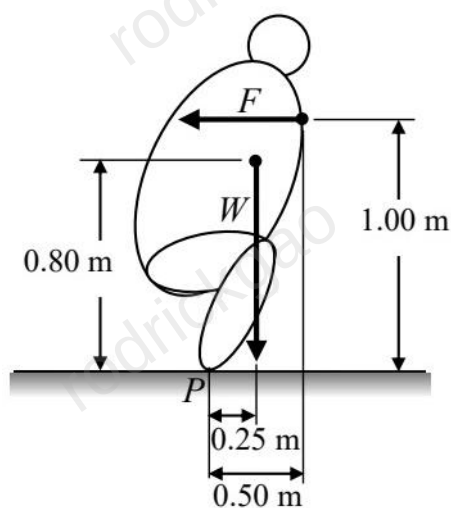
- A. the pulling force is acting internally within that person but it is not an external force acting on the person.
- B. it is impossible to generate a pulling force larger than a person's weight.
- C. the pulling force is balanced by the person's weight.
- D. it is impossible for the pulling force to overcome the force due to atmospheric pressure.

Official answer: A

For the whole person, the hand-hair pulling forces are internal and cancel in the total-force balance. They cannot provide an upward external impulse to lift the person's centre of mass. A force from an external object would be needed. Answer: A.

Question 10

A child tries to push a 150 kg (sumo) wrestler in order to topple him backwards in a game. The simplified diagram indicates the weight W of the wrestler and the horizontal pushing force F acting on the wrestler by the child. P is the wrestler's contact point on the ground. Find the minimum value of F required.



A. 368 N

B. 736 N

C. 1177 N

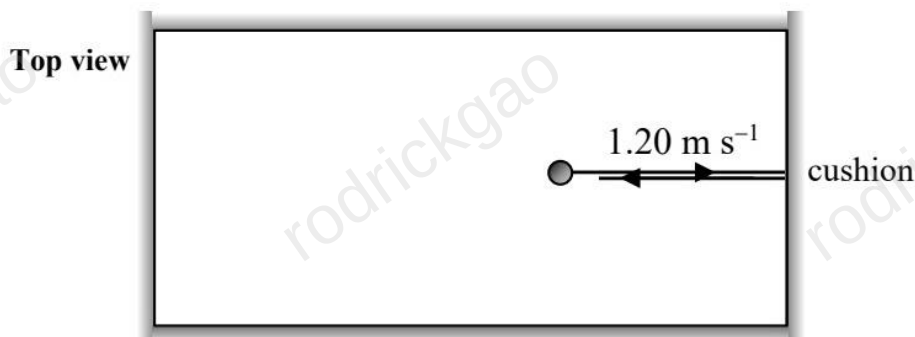
D. 2354 N

Official answer: A

At the threshold of toppling, take moments about the ground contact P . $F \times 1.00 = (150 \times 9.81) \times 0.25$, giving $F = 367.875 \text{ N} \approx 368 \text{ N}$. Use the perpendicular distances from P to the lines of action; the 0.80 m height of the centre of mass is not the moment arm of the vertical weight. Answer: A.

Question 11

On a billiard table, a billiard ball of mass 0.16 kg moving with a uniform speed of 1.20 m s^{-1} collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is 0.0003 s , find the average force acting on the ball by the cushion during the collision.



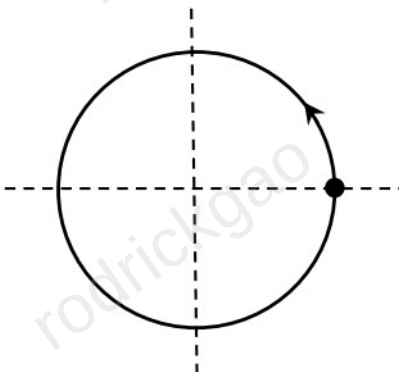
- A. 0 N
- B. 640 N
- C. 1280 N
- D. 2560 N

Official answer: C

The velocity reverses, so the magnitude of the momentum change is $\Delta p = 2mv = 2 \times 0.16 \times 1.20 = 0.384\text{ kg m s}^{-1}$. The average force is $\Delta p / \Delta t = 0.384 / 0.0003 = 1280\text{ N}$, directed away from the cushion. Unchanged speed does not mean unchanged momentum. Answer: C.

Question 12

The figure shows a particle performing uniform circular motion. Which of the following correctly indicates the directions of its velocity and acceleration at the instant shown?



- A. Velocity upward; acceleration left
- B. Velocity upper-left; acceleration left
- C. Velocity upper-left; acceleration right
- D. Velocity upward; acceleration right

Official answer: A

Velocity is tangential to the circular path, hence upward at the rightmost point for the indicated anticlockwise motion. Acceleration is centripetal, hence horizontally left towards the centre. Constant speed does not mean zero acceleration because the velocity direction changes. Answer: A.

Question 13

A satellite orbits the Earth in a circular orbit of radius r with a period of 24 hours. The period of another satellite which orbits the Earth in a circular orbit of radius $2r$ is

- A. less than 24 hours.
- B. 24 hours.
- C. between 24 hours and 48 hours.
- D. more than 48 hours.

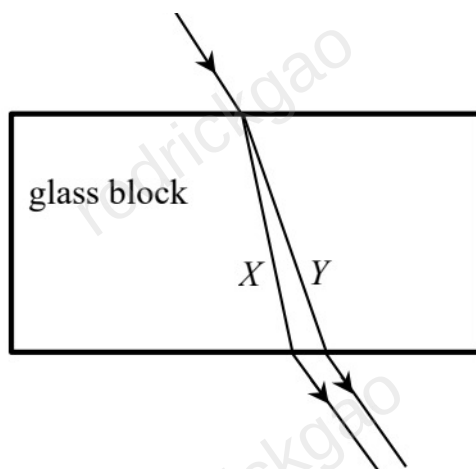
Official answer: D

For circular orbits about the same central mass, $GM/r^2 = 4\pi^2 r/T^2$, so $T^2 \propto r^3$. Doubling r multiplies T by $\sqrt[3]{8}$. The new period is $24\sqrt[3]{8} \approx 67.9$ h, which exceeds 48 h. Answer: D.

Question 14

A light beam consisting of monochromatic light beams X and Y is incident on a rectangular glass block. It splits into two beams as shown. Which of the following comparisons of beams X and Y is/are correct?

- (1) X has a higher frequency.
- (2) X has a larger speed in glass.
- (3) In the glass block, the critical angle for X is greater.



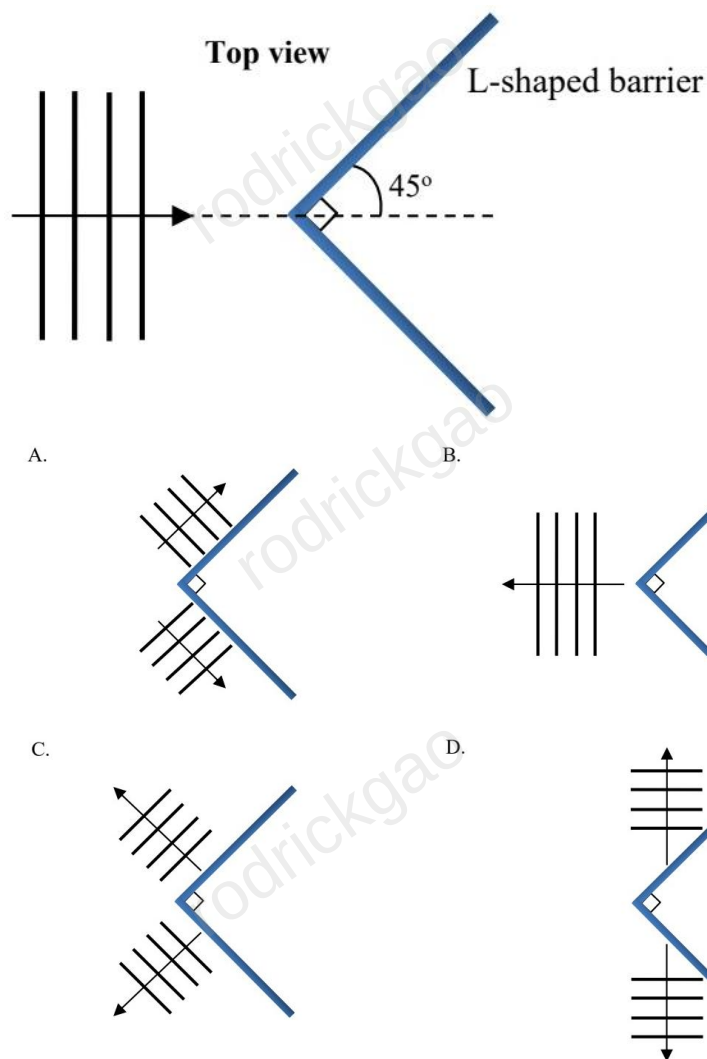
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

X bends more towards the normal, so $n_X > n_Y$. For ordinary dispersion in glass, this means X has the higher frequency. Since $v = c/n$ and $\sin c_{\text{critical}} = 1/n$, X has the smaller speed and smaller critical angle. Only statement (1) is correct. Answer: A.

Question 15

In a ripple tank, a series of plane water waves approach an L-shaped barrier as shown above. Which diagram below best shows the wave pattern reflected from the barrier?

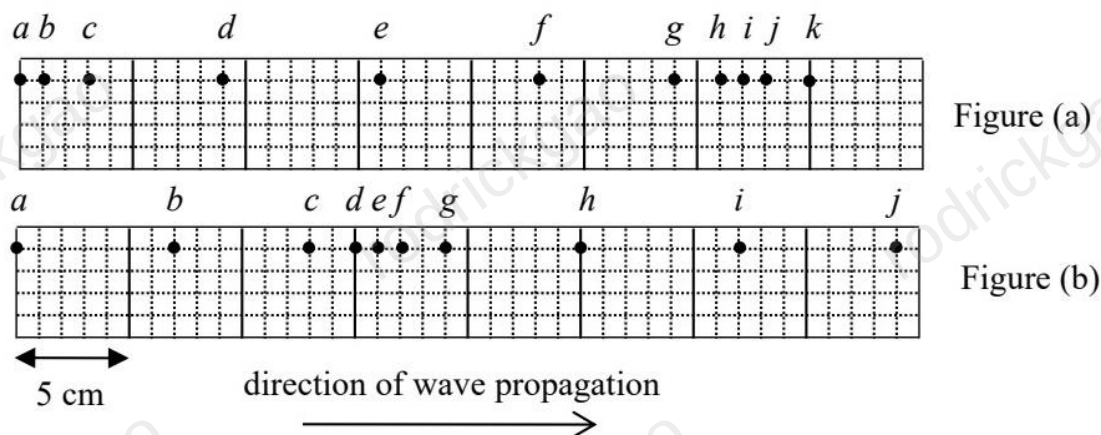


Official answer: D

Each face makes 45° with the incident direction. Applying angle of reflection = angle of incidence sends the upper reflected wave upward and the lower reflected wave downward. Their wavefronts are perpendicular to those directions and therefore horizontal. The wave speed and wavelength remain unchanged in the same water.
Answer: D.

Question 16

A series of particles is uniformly distributed along a slinky spring initially. Figures (a) and (b) show their positions at two instants when a travelling wave of frequency 10 Hz propagates along the spring from left to right. Which statement below is correct?



- A. Particle a is always stationary.
- B. Particles b and c vibrate with different amplitudes.
- C. The wavelength of the wave is 16 cm.
- D. The smallest time interval for obtaining these two figures is 0.05 s.

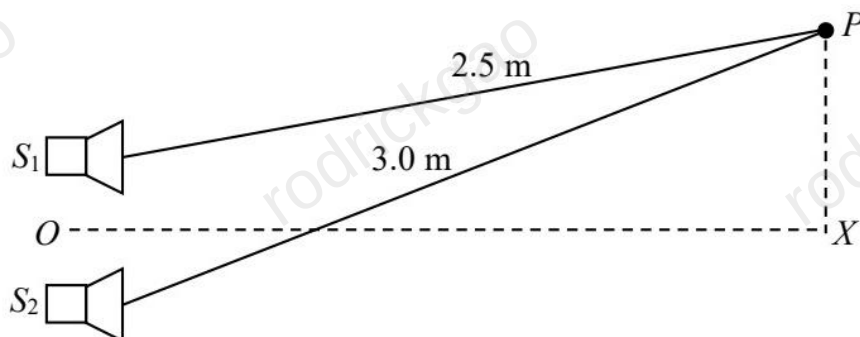
Official answer: D

The compression and rarefaction pattern interchanges between the two snapshots, corresponding to a phase change of π . The shortest interval is half a period: $\Delta t = T/2 = 1/(2f) = 1/(2 \times 10) = 0.05$ s. A particle can pass its equilibrium position in both snapshots without remaining stationary in between. Answer: D.

Question 17

Two identical loudspeakers S_1 and S_2 connected to a signal generator produce sounds of wavelength 0.10 m which are in anti-phase.

OX is the perpendicular bisector of the line joining S_1 and S_2 . What type of interference occurs at X and P respectively?



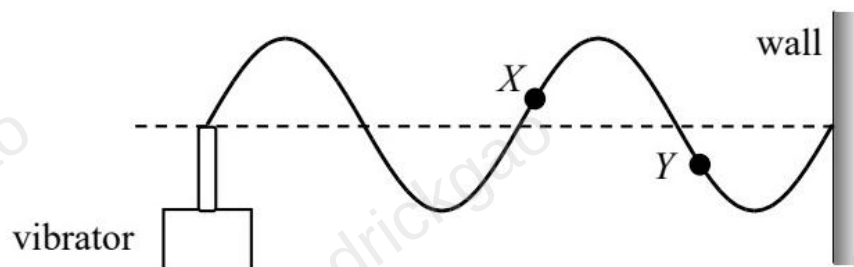
- A. At X : destructive; at P : constructive
- B. At X : constructive; at P : constructive
- C. At X : destructive; at P : destructive
- D. At X : constructive; at P : destructive

Official answer: C

At X the path difference is zero, so the initially antiphase waves remain antiphase and interfere destructively. At P the path difference is $3.0 - 2.5 = 0.50\text{ m} = 5\lambda$, adding a whole number of cycles and leaving them antiphase. Both positions therefore have destructive interference. Answer: C.

Question 18

A string is tied to a vibrator at one end while the other end is fixed to a wall. The figure shows the stationary wave formed at a certain frequency of the vibrator. Which statement is correct when the frequency is doubled?



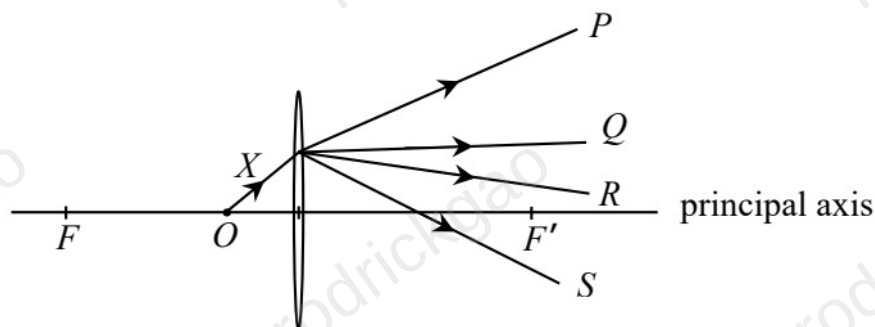
- A. The amplitude of the wave will double.
- B. The wavelength will double.
- C. The speed of the wave will double.
- D. Particle X and Y will become in phase.

Official answer: D

The wave speed is fixed by the string's tension and mass per unit length. Doubling f therefore halves λ and doubles the number of loops. X and Y then lie in loops whose oscillations have the same phase. The amplitude is not fixed by this frequency change alone. Answer: D.

Question 19

X is a light ray from an object O placed on the principal axis of a convex lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray?



- A. P
- B. Q
- C. R
- D. S

Official answer: A

O is inside the focal length of a convex lens. The emergent rays still diverge, although less strongly than the incident rays, and their backward extensions meet at a virtual image on the object side. Ray P has the required upward divergence; Q would correspond to the object at the focus, while R and S converge towards the axis. Answer: A.

Question 20

An object of height 10 cm placed at 20 cm in front of a concave lens forms an image of height 8 cm. Find the focal length of the concave lens.

- A. 4.4 cm
- B. 8.9 cm
- C. 40 cm
- D. 80 cm

Official answer: D

The upright image has magnification $8/10 = 0.8$. With a real object distance $u = 20$ cm, the virtual image distance is $v = -0.8u = -16$ cm. Then $1/f = 1/u + 1/v = 1/20 - 1/16 = -1/80 \text{ cm}^{-1}$, so $f = -80$ cm. The listed focal-length magnitude is 80 cm. Answer: D.

Question 21

Which description below is INCORRECT?

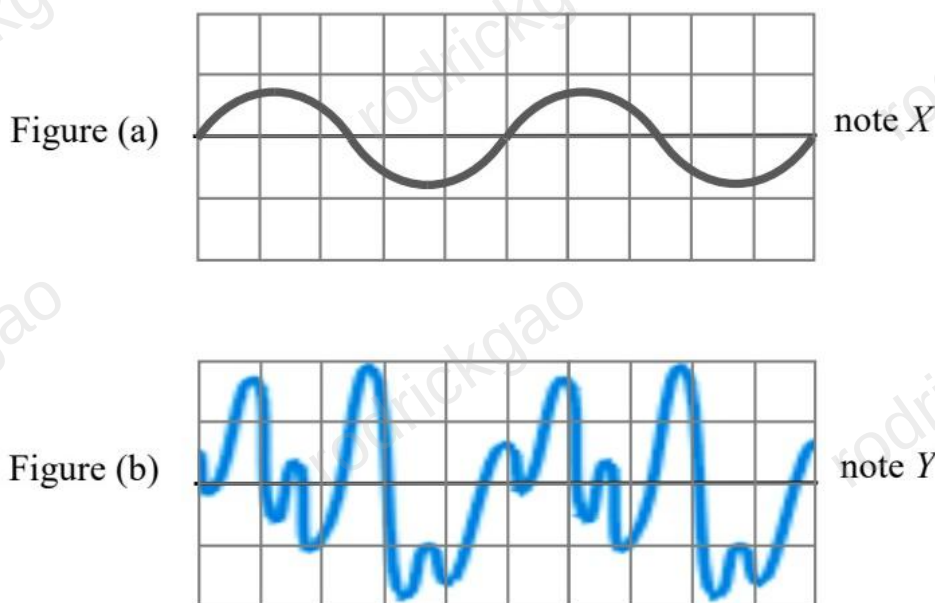
- A. Microwaves are a kind of electromagnetic waves.
- B. Microwaves can be observed with the naked eye.
- C. Microwaves propagate with the speed of light in a vacuum.
- D. Microwaves are used in radar to detect the position of an aeroplane.

Official answer: B

Microwaves are electromagnetic waves and travel at c in a vacuum. They are used in radar, but their wavelengths lie outside the visible band and they cannot be seen by the naked eye. The incorrect description is B. Answer: B.

Question 22

Figures (a) and (b) are the CRO displays of musical notes X and Y using the same time axis and intensity axis. Which comparison of these two notes must be correct?



- A. Pitch: the same; loudness: the same
- B. Pitch: the same; loudness: not the same
- C. Pitch: higher for Y; loudness: the same
- D. Pitch: higher for Y; loudness: not the same

Official answer: B

Each complete waveform repeats over the same horizontal interval, so X and Y have the same fundamental period, frequency and pitch. Y has a larger amplitude on the same vertical scale and therefore a different loudness. Extra peaks within one repeating waveform are harmonics, not additional fundamental cycles. Answer: B.

Question 23

Charges $+Q$ and $-2Q$ are fixed at X and Y as shown. Which descriptions about the electric field along the dotted line are correct?

- (1) Between X and Y, the respective electric fields due to the two charges are in the same direction.
- (2) To the right of Y, the resultant electric field is always directed to the left.
- (3) On the dotted line, there exist two points where the resultant electric field is zero.



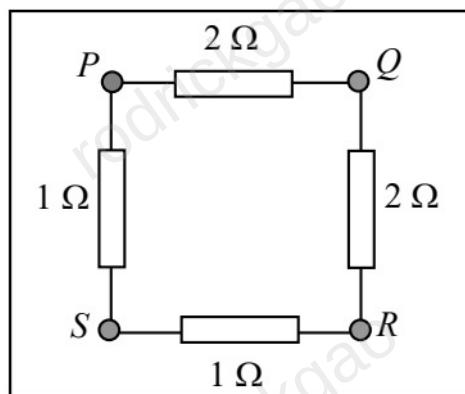
- A. (2) only
 B. (3) only
 C. (1) and (2) only
 D. (1) and (3) only

Official answer: C

Between X and Y, both fields point from $+Q$ towards $-2Q$, so (1) is true. To the right of Y, the nearer negative charge also has twice the magnitude, so its leftward field always dominates: (2) is true. Cancellation can occur only to the left of the smaller charge $+Q$, at one point, not two. Thus (3) is false. Answer: C.

Question 24

Four resistors are connected to terminals PQRS of a circuit board as shown. Across which pair of terminals below is the equivalent resistance smallest?



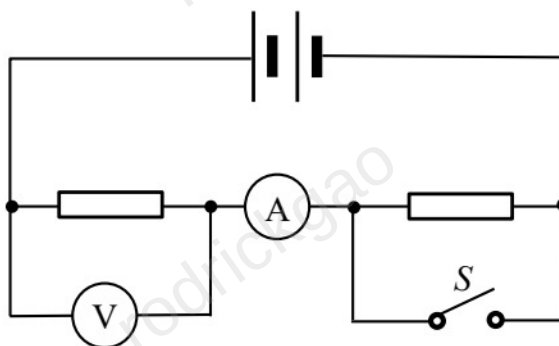
- A. PQ
 B. PR
 C. PS
 D. QS

Official answer: C

The two paths between any chosen terminals are in parallel. $R_{PQ} = 2 \times 4 / (2 + 4) = 4/3 \, \Omega$; $R_{PR} = 4 \times 2 / (4 + 2) = 4/3 \, \Omega$; $R_{PS} = 1 \times 5 / (1 + 5) = 5/6 \, \Omega$; $R_{QS} = 3 \times 3 / (3 + 3) = 3/2 \, \Omega$. The smallest value is across PS. Answer: C.

Question 25

In the circuit below, the two resistors are identical. When S is closed, how would the readings of the ammeter and voltmeter change?



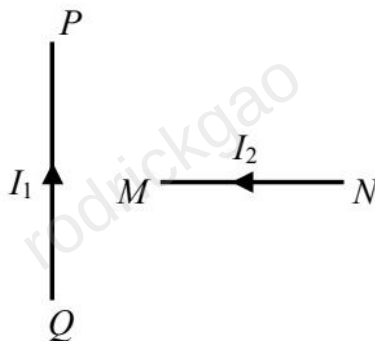
- A. Ammeter: increase; voltmeter: increase
- B. Ammeter: increase; voltmeter: decrease
- C. Ammeter: decrease; voltmeter: increase
- D. Ammeter: decrease; voltmeter: decrease

Official answer: A

Closing S short-circuits the right-hand resistor. The total resistance falls from $2R$ to R for ideal meters, so the current through the ammeter increases. The voltmeter is across the left-hand resistor, whose voltage IR also increases, from half the supply voltage to the full supply voltage for an ideal source. Answer: A.

Question 26

PQ in the figure is a section of a fixed long straight wire carrying a current I_1 . MN is a section of another long straight wire carrying a current I_2 . MN is perpendicular to PQ and both wires are in the same plane. What is the direction of the magnetic force acting on MN due to PQ? Neglect the effects of the Earth's magnetic field.



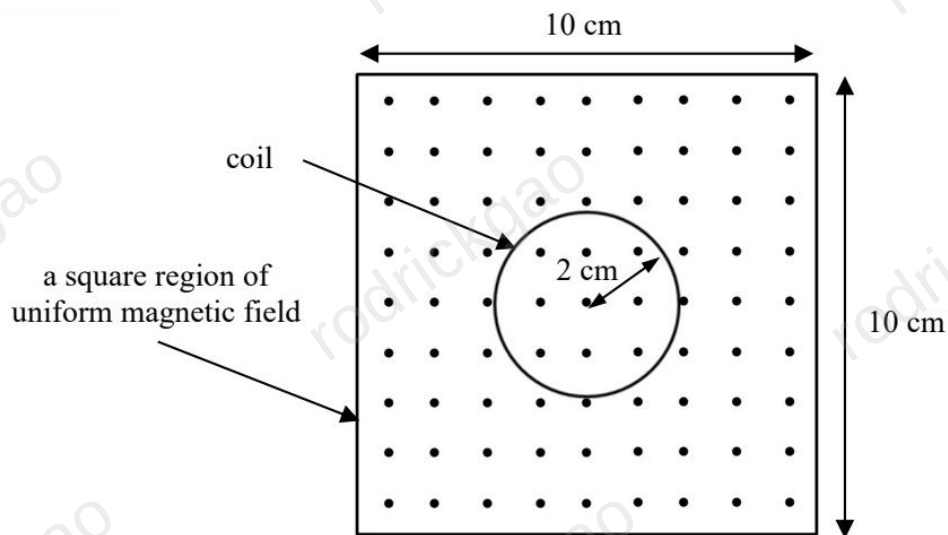
- A. pointing out of the paper
- B. downwards
- C. pointing into the paper
- D. upwards

Official answer: B

The upward current in PQ produces a magnetic field into the paper at MN, by the right-hand grip rule. Current in MN is towards the left. Applying Fleming's left-hand rule, or the direction of $I\ell \times B$, gives a downward force on MN. Answer: B.

Question 27

A coil of radius 2 cm is placed inside a square region of uniform magnetic field pointing out of the paper as shown.



The magnetic field decreases uniformly from a strength of 5.0 T to 3.0 T in 0.8 s. What is the magnitude of the e.m.f. induced in the coil?

- A. 2.5×10^{-2} V
- B. 3.1×10^{-2} V
- C. 3.1×10^{-3} V
- D. 6.2×10^{-3} V

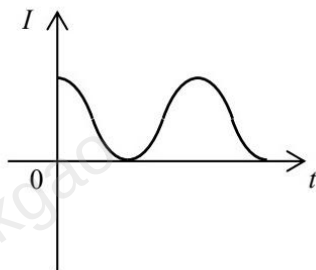
Official answer: C

Faraday's law for the single loop gives $|\mathcal{E}| = A|\Delta B|/\Delta t$. With $r = 0.020$ m, $|\mathcal{E}| = \pi(0.020)^2 \times (5.0 - 3.0)/0.8 = 3.14 \times 10^{-3}$ V. The radius must be converted to metres before squaring. Answer: C.

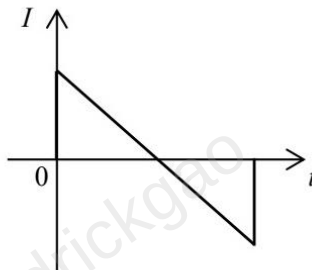
Question 28

Which of the following current-against-time (I - t) graphs represent an a.c.?

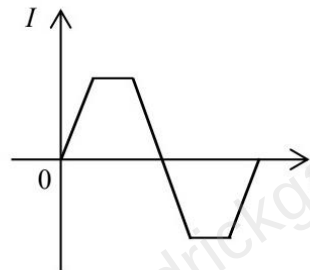
(1)



(2)



(3)



A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

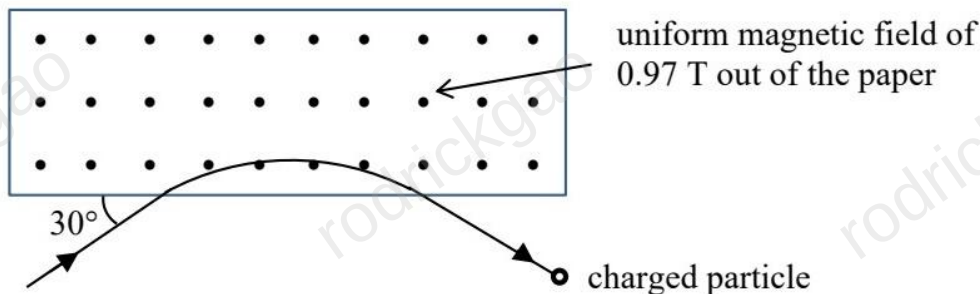
D. (1), (2) and (3)

Official answer: C

An alternating current changes direction, so its graph must contain both positive and negative current values. Graphs (2) and (3) do so. Graph (1) changes magnitude but never reverses direction and is therefore a varying direct current. An a.c. need not be sinusoidal. Answer: C.

Question 29

A charged particle enters and leaves a magnetic field of uniform flux density 0.97 T along the path as shown. It experiences a magnetic force of $5.9 \times 10^{-12} \text{ N}$ within the magnetic field. Given that the speed of the particle is $1.9 \times 10^7 \text{ m s}^{-1}$, find its charge.



- A. $-3.2 \times 10^{-19} \text{ C}$
- B. $+3.2 \times 10^{-19} \text{ C}$
- C. $-6.4 \times 10^{-19} \text{ C}$
- D. $+6.4 \times 10^{-19} \text{ C}$

Official answer: B

The velocity lies in the paper while B points out of the paper, so v is perpendicular to B throughout the field. $|q| = F/(vB) = (5.9 \times 10^{-12}) / (1.9 \times 10^7 \times 0.97) \approx 3.2 \times 10^{-19} \text{ C}$. The clockwise curvature is the direction of $v \times B$ for a positive charge. The 30° angle to the boundary is not the angle between v and B . Answer: B.

Question 30

A power station generates 217 MW of power for transmission to downtown under a voltage of 132 kV via a transmission cable which has an effective resistance of $0.02 \Omega \text{ km}^{-1}$. The power loss for a total cable length of 40 km is about

- A. 2.16 MW .
- B. 6.49 MW .
- C. 13.0 MW .
- D. 21.6 MW .

Official answer: A

The transmission current is $I = P/V = 217 \times 10^6 / (132 \times 10^3) = 1.64 \times 10^3 \text{ A}$. The resistance of the stated total cable length is $R = 0.02 \times 40 = 0.80 \Omega$. Hence the loss is $I^2 R = 2.16 \times 10^6 \text{ W} = 2.16 \text{ MW}$. Do not double the cable length again. Answer: A.

Question 31

Which statements about domestic electricity are correct?

- (1) All fuses should be installed in the live wire.
- (2) An electrical appliance can work even without an earth wire.
- (3) An electrical appliance having a fuse in it can protect people from getting an electric shock.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

A fuse belongs in the live wire so that blowing it disconnects the appliance from the live supply. An appliance can operate without an earth wire because its normal operating circuit is live–neutral, though earthing is essential where required for safety. A fuse alone does not guarantee protection from electric shock; a harmful body current can be too small to blow it. Statements (1) and (2) are correct. Answer: A.

Question 32

It takes 2.0 years for the activity of a sample containing a radioactive nuclide to drop from 800 Bq to 600 Bq. How long will it take for the activity of the sample to drop from 600 Bq to 300 Bq?

- A. 2.0 years
- B. 2.5 years
- C. 3.0 years
- D. 4.8 years

Official answer: D

Use $A = A_0 e^{-kt}$. From $600/800 = e^{-2k}$, $k = \ln(4/3)/2 \text{ year}^{-1}$. The fall from 600 to 300 Bq is one half-life: $t = \ln 2/k = 2 \ln 2 / \ln(4/3) = 4.82 \text{ years}$. Activity does not fall by equal amounts in equal times. Answer: D.

Question 33

In order to locate the cracks in an underground water pipe, a small amount of a radioisotope is added to the water in the pipe. The radioactive substance will leak through the crack to the soil nearby, thus radiation will be detected above ground. Which radioisotope (P, Q, R or S) below is the most suitable for this purpose?

- A. Radioisotope; half-life; radiation emitted: P — 66 hours — β
- B. Radioisotope; half-life; radiation emitted: Q — 36 hours — γ
- C. Radioisotope; half-life; radiation emitted: R — 7.2 hours — α
- D. Radioisotope; half-life; radiation emitted: S — 45 minutes — α, β

Official answer: B

The tracer must emit radiation that penetrates the pipe and soil to reach an above-ground detector. Gamma radiation from Q is suitable, whereas alpha and beta particles are much more readily absorbed. Its 36-hour half-life also allows the investigation without remaining strongly active for an unnecessarily long time. Answer: B.